
When Averaged Field Regularity Fails to Rank Few-Step Generative Paths

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Abstract

1 Choosing a probability path for few-step generative sampling is often guided by
2 averaged regularity of the velocity field: a path with smaller averaged squared Jaco-
3 bian norm is expected to discretize better at a fixed number of function evaluations
4 (NFE). We test this assumption where every quantity is exact: commuting Gaussian
5 probability flows, whose numerical endpoint laws and Gaussian 2-Wasserstein
6 (W_2) errors are closed-form. The averaged-regularity ordering of linear versus
7 variance-preserving paths fails to determine the fixed-NFE W_2 ordering: fourteen
8 ranking inversions reproduce across Euler, Heun, and RK4, survive an 80-digit
9 precision audit, and recur in a pre-registered non-centered family. An exact modal
10 decomposition explains the mismatch through solver-stage sampling and signed
11 one-step defects transported to the endpoint, and an explicit scalar construction
12 proves the non-implication for fixed-grid Euler. This is a controlled limitation
13 study: it does not refute regularity-guided schedule design, and no learned model
14 is evaluated.

15 1 Introduction

16 Few-step sampling of continuous-time generative models commits to a probability path, a solver, and
17 a small NFE budget before any error is observed. A natural criterion, made explicit by Lipschitz-
18 guided schedule design [Chen et al., 2025], ranks candidate paths by the averaged squared Jacobian
19 (Lipschitz) norm of the velocity field. Averaged regularity controls worst-case error *bounds* [Hairer
20 et al., 1993], but bounding an error is weaker than determining which of two paths errs less at an equal
21 small budget. We isolate that question where the marginal flow is affine, the fixed-step endpoint law
22 is exactly computable, and the error is the closed-form Gaussian W_2 [Gelbrich, 1990]: no sampling,
23 training, or estimation noise intervenes, and every reported digit is auditable.

24 Contributions.

- 25 • A controlled exact- W_2 benchmark on commuting Gaussian flows in which the averaged-regularity
26 ordering fails to determine the fixed-NFE W_2 ordering in 14 of 36 pre-specified comparisons,
27 verified against an 80-digit reference and reproduced (11 of 18 comparisons) in a pre-registered
28 non-centered family frozen before execution.
- 29 • An exact eigenmode analysis showing the endpoint factor error is a transported sum of signed,
30 solver-stage-dependent local defects—information an unsigned time average of Jacobian norms
31 cannot retain.
- 32 • An explicit smooth scalar construction proving that, for every $L > 0$ and grid size N , strictly
33 larger averaged squared Jacobian is compatible with zero fixed-grid Euler endpoint error.

34 Solver dependence of good schedules is not itself novel [Sabour et al., 2024, Karras et al., 2022]; the
35 contribution is the exact, controlled non-implication and its modal explanation. Averaged regularity

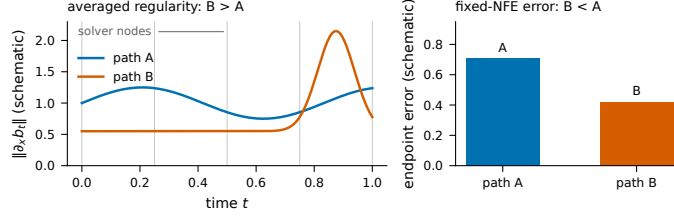


Figure 1: **Conceptual schematic (not data).** Path B has the larger time-averaged Jacobian magnitude, but its irregularity concentrates between the nodes a fixed-NFE solver samples; the endpoint-error ordering can oppose the averaged-regularity ordering.

may well remain useful for schedule *design*; our result bounds what it can certify about fixed-budget *rankings* in the tested systems.

2 Gaussian probability flows and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent and let a scalar schedule $(\alpha(t), \sigma(t))$, with $\alpha(0) = \sigma(1) = 1$ and $\alpha(1) = \sigma(0) = 0$, define the interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023]. The marginal law is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q(t) = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$, and the marginal velocity field is affine,

$$b(t, x) = A(t)x + c(t), \quad A(t) = C_t Q(t)^{-1}, \quad c(t) = \dot{m}_t - A(t)m_t, \quad (1)$$

with $C_t = \dot{\alpha}\alpha\Sigma_0 + \dot{\sigma}\sigma\Sigma_1$; centered with $\Sigma_0 = I$ this is $A(t) = \frac{1}{2}Q'(t)Q(t)^{-1}$, $c \equiv 0$. We compare the *linear* path $(1-t, t)$ and the *variance-preserving* (VP) path $(\cos \frac{\pi t}{2}, \sin \frac{\pi t}{2})$ [Song et al., 2021]. Following Chen et al. [2025], the averaged-regularity baseline is $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2 dt$ (exact spectral norms, fixed trapezoidal grid); note $\mathcal{R}$ ignores $c(t)$.

A fixed-step solver applied to (1) induces an affine endpoint map, recovered exactly from $d+1$ probe trajectories; pushing the source moments through it and applying the Gaussian W_2 formula [Gelbrich, 1990] gives the exact endpoint error. With $\Sigma_0 = I$ and $\Sigma_1 = U \text{diag}(\lambda_i) U^\top$ all matrices commute: each mode obeys $x'_i = a_i(t)x_i + c_i(t)$ with $a_i = q'_i/(2q_i)$, $q_i = \alpha^2 + \sigma^2\lambda_i$, and in the centered case $W_2^2 = \sum_i (|r_i| - \sqrt{\lambda_i})^2$, where r_i is the product of the numerical one-step factors of mode i .

3 Why the ranking need not agree: transported signed defects

The exact one-step transition of mode i over $[t_n, t_{n+1}]$ is $e_{i,n} = \sqrt{q_i(t_{n+1})/q_i(t_n)}$, while a Runge-Kutta step multiplies by a factor $r_{i,n}$ built from stage samples of a_i (one for Euler, two for Heun, four for RK4). For smooth a , the leading local defect (exact minus method) is $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-\frac{1}{12}a'' + \frac{1}{6}a^3)h^3$ for Heun—different derivatives, signs, and stage locations. The endpoint factor error obeys the exact telescoping identity

$$\prod_{n < S} r_{i,n} - \prod_{n < S} e_{i,n} = \sum_{n < S} (r_{i,n} - e_{i,n}) \left(\prod_{j < n} r_{i,j} \right) \left(\prod_{j > n} e_{i,j} \right), \quad (2)$$

with no smallness or positivity assumption. This identifies what the scalar $\mathcal{R}$ discards: which times the stages sample, the derivatives of A , the *signs* of local defects, their cancellation, and their transport to the endpoint. An unsigned average of $\|A\|^2$ retains none of these, so it need not determine the fixed-NFE ordering.

4 Controlled benchmark and results

Design. Source $N(0, I)$; anisotropic (condition number 4) and seeded low-rank $(FF^\top + 0.05I, \text{rank } 2)$ targets; $d \in \{2, 8\}$; linear and VP paths; Euler, Heun, RK4; NFE $\in \{8, 16, 32\}$ with equal-NFE accounting (steps = NFE/stages): 72 float64 CPU configurations in 36 two-path blocks. A *ranking*

Table 1: Strongest inversion: low-rank Gaussian, $d = 8$, Euler, NFE 8. (phase4_gaussian_reproduction_2026-07-24-v1:results)

path	$\mathcal{R}[b]$ (lower = preferred)	Gaussian W_2
linear	2.9476523251	0.8108540111
variance-preserving	4.7295206355	0.4564779075

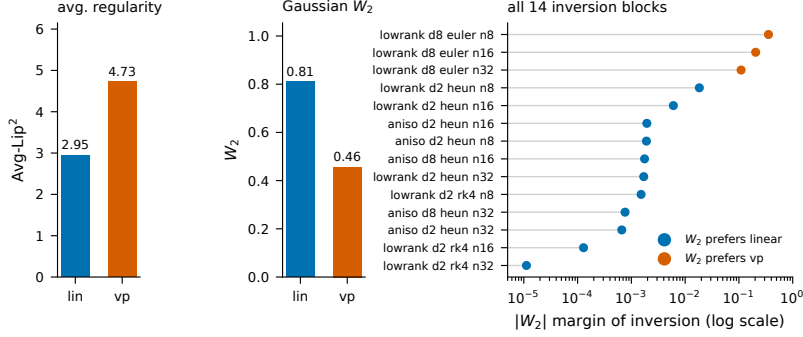


Figure 2: **Left:** the strongest inversion in both quantities. **Right:** W_2 margins of all fourteen inversion blocks (log axis; compresses, never exaggerates); color marks the path exact error prefers, always opposite to the baseline. (phase4_gaussian_reproduction_2026-07-24-v1:results)

66 *inversion* is a block where the $\mathcal{R}$ and W_2 orderings strictly disagree. Grid, definitions, and tolerances
67 were fixed before execution; post-hoc diagnostics are labeled as such.

68 **Fourteen audited inversions.** All 72 configurations reproduce a registered
69 earlier run within 10^{-12} , and 14 of 36 blocks are inversions (Figure 2;
70 phase4_gaussian_reproduction_2026-07-24-v1:results). The strongest is the low-rank
71 target, $d = 8$, Euler, NFE 8 (Table 1): the baseline prefers linear while its W_2 is 78% larger (margin
72 0.3543761036). All 72 float64 W_2 values agree with an 80-digit mpmath modal reference to at most
73 9.71×10^{-10} ; the smallest surviving inversion margin, 1.1189×10^{-5} , exceeds this by a factor
74 above 11,500 (phase4_precision_2026-07-24-v1:table).

75 **Solver-path interaction.** In the low-rank grid, VP is preferred under Euler and linear under
76 Heun and RK4, at both dimensions and every primary budget (Figure 3); the pattern persists at
77 NFE 64 and 128 and under $\pm 10\%$ target perturbations (phase4_robustness_2026-07-24-v1:table).
78 Solver-dependent schedule preference is established [Sabour et al., 2024]; the point is that a solver-
79 independent scalar cannot represent it. A solver-specific leading proxy from Section 3 agrees with 29
80 of 36 observed preferences versus 22 of 36 for the baseline, but this is post-hoc and in-sample and
81 carries no predictive claim (phase4_diagnostics_2026-07-24-v1:table).

82 **Pre-registered external validation.** To leave the original design space we froze, before execution,
83 one non-centered family: source $N(\mu_0, I)$, $\mu_{0,i} = 0.75(-1)^i$; target $N(\mu_1, D)$, $\mu_{1,i} = 1 + 0.25i$, D
84 geometric with condition number 6—so $c(t) \neq 0$ in (1)—with the same paths, solvers, and budgets.
85 Eleven of eighteen blocks are inversions ($\mathcal{R}$ prefers VP, margin 0.7513; exact W_2 prefers linear),
86 and all eleven pass the 80-digit audit (max float64 deviation 2.07×10^{-11} ; smallest inverted margin
87 1.42×10^{-6}) (workshop_external_validation_2026-07-24-v1:results). This supports the finding
88 only for the tested systems. No mixture target supports any conclusion here; mixture evidence was
89 excluded after an estimator-calibration failure.

90 5 A minimal non-implication for fixed-grid Euler

91 The benchmark shows the failure empirically; the following shows the implication already fails
92 logically, in one smooth scalar dimension.

93 **Proposition 1.** Fix $L > 0$ and an integer $N \geq 1$; let $h = 1/N$ and let forward Euler use the
94 left-endpoint grid $t_n = nh$ on $[0, 1]$. Set $\epsilon_N = N\{e^{L/N} - 1\} - L > 0$ and define $f_k(t, x) = a_k(t)x$

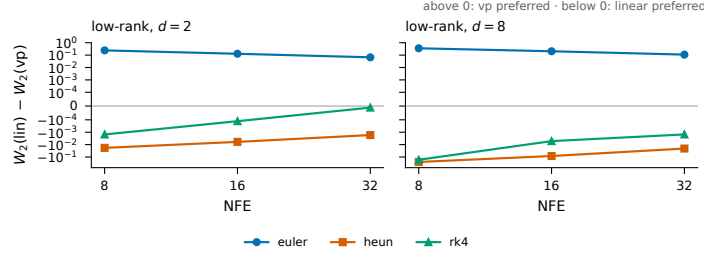


Figure 3: Signed low-rank margins $W_2(\text{lin}) - W_2(\text{vp})$ across solvers and NFE (symmetric log axis): the preferred path flips with the solver at equal NFE. (phase4_gaussian_reproduction_2026-07-24-v1:results)

95 with $a_0(t) = L$ and $a_{1,N}(t) = L + \epsilon_N \cos(2\pi Nt)$. For $x(0) = 1$: (i) both exact endpoints equal
 96 e^L ; (ii) $\int_0^1 a_{1,N}^2 = L^2 + \epsilon_N^2/2 > \int_0^1 a_0^2$; (iii) Euler is exact for $a_{1,N}$; (iv) Euler has strictly positive
 97 endpoint error for a_0 . Hence larger averaged squared Jacobian does not imply larger fixed-grid
 98 Euler endpoint error.

99 *Proof.* (i) $\int_0^1 \cos(2\pi Nt) dt = 0$ for integer N , so both coefficients integrate to L and $x(1) = e^L$. (ii)
 100 By orthogonality $\int_0^1 a_{1,N}^2 = L^2 + \epsilon_N^2/2$, and $\epsilon_N > 0$ since $e^z > 1 + z$ for $z = L/N > 0$. (iii) At
 101 every node $\cos(2\pi Nt_n) = 1$, so each Euler factor is $1 + h(L + \epsilon_N) = e^{L/N}$ and the product is e^L .
 102 (iv) $(1 + L/N)^N < e^L$ since $\log(1 + z) < z$ for $z > 0$. $\square$

103 The quantifiers matter: $a_{1,N}$ is chosen after the grid, its amplitude vanishes as $N \rightarrow \infty$, and shifted
 104 grids or multi-stage solvers sample different phases. This elementary grid-aliasing construction,
 105 which passed an independent adversarial review, establishes a logical non-implication for fixed-grid
 106 left-endpoint Euler only; we claim neither the mechanism behind every Gaussian inversion nor a new
 107 numerical-analysis phenomenon.

108 6 Related work

109 Flow matching and stochastic interpolants define the Gaussian path family and its marginal fields [Lip-
 110 man et al., 2023, Albergo et al., 2025, Liu et al., 2023]; Lipschitz-guided schedule design proposes
 111 the criterion we test [Chen et al., 2025]. Fast diffusion solvers exploit exactly integrated linear parts
 112 and solver-specific expansions [Lu et al., 2022a,b, Zhang and Chen, 2023], EDM separates schedule
 113 from discretization [Karras et al., 2022], and Align Your Steps optimizes schedules per solver, model,
 114 and dataset [Sabour et al., 2024]: schedule quality is known to be solver-dependent. Runge–Kutta
 115 and backward error analysis supply the local-error machinery [Butcher, 1963, Hairer et al., 1993,
 116 2006]; Gaussian W_2 is due to Gelbrich [1990]. None of these states the exact commuting inversion
 117 benchmark or the grid-aware non-implication, though search absence does not establish novelty.

118 7 Limitations and conclusion

119 Our systems are exact, commuting, affine Gaussian flows with two paths, three solvers, dimensions
 120 2 and 8, and small budgets, plus one pre-registered non-centered family. No learned neural field is
 121 evaluated; no image, video, or world-model improvement is demonstrated or implied; the solver-
 122 specific proxy is post-hoc and in-sample; the proposition is grid-aware by construction. Averaged
 123 regularity may remain valuable for schedule design, and our result does not refute Lipschitz-guided
 124 schedules: in the tested systems the averaged-regularity *ordering* does not determine the fixed-NFE
 125 W_2 ordering, because fixed-budget error lives in solver-stage samples and transported signed defects
 126 invisible to an unsigned time average. The benchmark is a small, auditable test bed for schedule-
 127 selection criteria: every numerical statement carries a checksummed artifact identifier whose manifest
 128 records commit, config and environment hashes, and seeds; the anonymized audit bundle accompanies
 129 submission.

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